

SUGAR LAND, TX, USA

WMO: 722543

Lat: 29.622N

Lon: 95.657W

Elev: 26

StdP: 101.02

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	1.4	-7.9	1.9	6.1	-5.8	2.3	7.4	10.9	15.4	9.8	16.4	2.6	0	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	36.3	24.9	35.1	25.0	34.1	25.1	26.9	31.5	26.6	31.2	26.2	30.9	3.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.5	28.5	25.5	20.7	28.3	25.1	20.2	28.2	84.6	31.4	82.7	31.0	81.3	30.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.4	7.7	-3.6	38.0	2.1	1.6	-5.1	39.1	-6.3	40.0	-7.5	40.9	-9.0	42.0		
(5)			-4.8	27.6	2.0	0.6	-6.3	28.1	-7.5	28.4	-8.7	28.8	-10.1	29.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	12.0	14.4	17.8	21.3	25.1	28.1	28.8	29.1	26.9	22.3	17.1	13.3
	DBStd	7.19	5.21	5.47	4.58	3.55	2.79	1.66	1.47	1.54	2.23	4.05	5.06	5.29
	HDD10.0	94	38	21	4	0	0	0	0	0	0	0	6	26
	HDD18.3	687	207	131	67	14	1	0	0	0	0	13	82	172
	CDD10.0	4248	99	143	244	339	469	543	584	593	506	381	219	128
	CDD18.3	1800	9	19	49	103	212	293	326	334	256	136	45	17
	CDH23.3	18062	31	74	239	683	1924	3303	3790	3943	2529	1199	277	69
	CDH26.7	7652	1	4	28	140	688	1526	1804	1953	1066	393	46	3
Wind	WSAvg	3.4	3.7	4.0	4.1	4.2	4.0	3.3	2.8	2.7	2.6	3.0	3.3	3.6
Precipitation	PrecAvg	1268	90	71	92	107	122	120	102	110	111	126	106	95
	PrecMax	1759	251	234	219	274	316	341	339	237	358	324	363	239
	PrecMin	445	3	8	13	1	1	6	19	10	23	0	5	19
	PrecStd	331	60	56	57	82	84	81	85	63	77	102	81	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.9	27.4	29.1	31.3	34.1	36.9	37.2	38.0	36.2	33.0	30.0	27.1
		MCWB	19.6	20.3	20.7	22.7	23.8	24.3	24.9	24.8	24.2	24.5	22.8	21.3
	2%	DB	24.0	25.6	27.3	29.4	32.7	35.1	35.9	36.5	34.4	31.9	27.9	25.3
		MCWB	18.9	19.6	20.6	22.0	23.8	24.6	24.9	25.2	24.6	24.2	21.8	20.6
	5%	DB	22.4	23.9	26.1	28.2	31.6	34.0	34.7	35.1	33.0	30.6	26.5	23.7
		MCWB	18.5	19.4	19.9	21.6	23.5	24.8	25.1	25.4	24.4	23.2	20.9	20.3
	10%	DB	20.9	22.5	24.4	27.2	30.5	32.9	33.6	34.0	32.0	28.9	24.8	22.3
		MCWB	17.6	18.9	19.3	21.0	23.3	24.9	25.2	25.4	24.3	22.1	20.5	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.2	22.8	24.7	26.1	27.4	27.2	27.7	27.0	26.2	24.5	23.1
		MCDB	23.7	24.9	26.2	29.1	30.4	31.4	31.8	32.2	31.1	29.8	27.4	24.8
	2%	WB	20.6	21.4	22.0	23.7	25.3	26.7	26.7	27.0	26.4	25.6	23.4	22.0
		MCDB	22.6	23.8	25.1	27.3	29.6	31.0	31.5	31.7	30.7	29.3	26.1	24.1
	5%	WB	19.7	20.6	21.4	23.0	24.8	26.1	26.3	26.6	25.9	25.0	22.5	21.2
		MCDB	21.8	22.9	24.3	26.3	29.0	30.6	31.2	31.4	30.2	28.6	25.1	23.3
	10%	WB	18.3	19.7	20.8	22.5	24.4	25.7	25.9	26.2	25.5	24.2	21.5	19.8
		MCDB	20.2	21.9	23.4	25.5	28.4	30.3	30.7	31.1	29.7	27.4	24.1	21.8
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.2	10.3	10.3	9.9	9.7	9.6	10.2	9.8	11.4	11.3	10.6
		MCDBR	12.0	10.7	10.4	10.4	10.5	11.0	11.2	11.7	11.1	10.8	11.2	11.1
		MCWBR	7.7	6.4	5.2	4.6	3.4	3.0	2.9	2.9	3.5	4.2	6.0	7.3
	5% WB	MCDBR	10.0	8.8	8.6	8.2	8.6	9.0	9.5	10.1	9.2	9.1	9.4	9.9
		MCWBR	7.5	6.5	5.2	4.5	3.4	3.2	3.1	3.2	3.3	4.5	6.3	7.8
Clear-Sky Solar Irradiance	taub	0.344	0.360	0.376	0.438	0.484	0.468	0.496	0.485	0.455	0.381	0.375	0.349	
	taud	2.482	2.450	2.395	2.223	2.129	2.228	2.169	2.201	2.253	2.447	2.413	2.473	
	Ebn at Noon	891	906	911	861	818	827	802	809	822	870	846	864	
	Edh at Noon	92	103	116	143	158	142	150	144	133	102	98	89	
All-Sky Solar Radiation	RadAvg	2.87	3.24	4.42	5.28	5.86	6.12	6.01	5.76	4.98	4.32	3.25	2.56	
	RadStd	0.28	0.57	0.38	0.39	0.43	0.49	0.57	0.46	0.52	0.51	0.34	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	+235	+169
(47) Regional (33 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+97

Nomenclature: See separate page